QPRC

QUEANBEYAN - PALERANG REGIONAL COUNCIL

DEVELOPMENT DESIGN SPECIFICATION

D6

SITE REGRADING

VERSION 1 – JANUARY 2019

AUS-SPEC #1

Amendment Record for this Specification Part

This Specification is Council's edition of the AUS-SPEC generic specification part and includes Council's primary amendments for development.

Details are provided below outlining the clauses amended from the Council edition of this AUS-SPEC Specification Part. The clause numbering and context of each clause are preserved. New clauses are added towards the rear of the specification part as special requirements clauses. Project specific additional script is shown in the specification as italic font.

The amendment code indicated below is 'A' for additional script 'M' for modification to script and 'O' for omission of script. An additional code 'P' is included when the amendment is project specific.

Amendment Sequence No.	Key Topic addressed in amendment	Clause No.	Amendment Code	Author Initials	Amendment Date
VERSION 1	Standards updated	D6.03	M	KD	19/04/10
	EPA guidelines referenced	D6.09.1	M		
	Addition of Batter Slopes Clause	D6.13	A	HS	19/08/2016
	Addition of Retaining Walls Clause	D6.14	A	HS	
	Addition of two specification objectives	D6.02	A	HS	
	Addition of one site grading concept	D6.04	A	HS	
	Max grade of lots added, requirements for lots steeper than 20%	D6.05.7	A	CS	21/12/2018

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GENERAL

D6.01 SCOPE

1. This Design Specification sets out requirements for the site regrading involved in land development and subdivision. Conceptual requirements are presented as necessary considerations when preparing designs for site regrading.

2. The scope of this Specification assumes that the Designer is familiar with requirements cited in the various construction specifications, specifically those related to earthworks, clearing and grubbing, erosion and sedimentation. Additionally the Designer needs to make reference to the associated design specifications related to stormwater drainage design, geometric road design and erosion control and stormwater management.

Familiarity with other Specifications Required

D6.02 OBJECTIVES

1. This Specification aims to assist the Designer in achieving:

- efficient and economical design
- enhancement of the environmental character of the site whilst maintaining the natural features of the site

Environmentally Sound

- provision of safe conditions for construction commensurate with the proposed purpose of the development

Safe for Construction

- equality of building conditions for residential development
- a minimal impact on adjoining properties and developments.

Impact on Adjoining Properties

- sites that are presented in a consistent manner and in accordance with the Development Control Plan.
- sites that provide prospective purchasers with clarity as to the extent of earthworks and or retaining structures that may be required as part of their dwelling construction.

D6.03 REFERENCE AND SOURCE DOCUMENTS

(a) Council Specifications

Construction Specifications

C211 - Control of Erosion and Sedimentation – Version 1
C212 - Clearing and Grubbing – Version 1
C213 - Earthworks – Version 1
C273 - Landscaping – Version 1

Design Specifications

D1 - Geometric Road Design
D5 - Stormwater Drainage Design
D7 - Erosion Control and Stormwater Management
D13 - Vehicular Access Design

(b) Australian Standards

AS 3798:2007 - Guidelines on earthworks for commercial and residential developments
AS 2870.1 - Residential slabs and footings - Construction.

D6.04 SITE REGRADING CONCEPT

1. Areas of a site proposed for building or recreational purposes may not be suitable in their natural state for their intended function without improvement works to:

(a) Alleviate flooding of low-lying ground

(b) Fill gullies or create emergency flowpaths after underground stormwater piping has been installed

(c) Allow improved runoff from flat ground

(d) Regrade excessively steep slopes that would preclude economical construction of dwelling foundations

(e) Allow effective recreational use or give reasonable access

(f) Provide economically feasible building sites for prospective purchasers

The Designer shall review the natural surface contours and where necessary shall design finished surface levels that ensure the land is suitably prepared

2. Where practical, areas should be regraded to minimise the necessity for underground drainage systems with surface inlet pits, and allow surface water to flow naturally to roads or drainage reserves without excessive concentration. ***Drainage***

3. The Designer shall consider the implications of site regrading in relation to the existing natural environment. Generally site regrading shall be minimised in heavily treed areas. ***Natural Environment***

4. Care shall be taken to provide depressions for overland flow from low points and over major drainage lines, to direct stormwater for storms up to a 100 year average recurrence interval (ARI). ***Overland Flow***

5. The design of site regrading areas in conjunction with the design of roadworks shall be considered with the objective of balancing cut to fill and achieving both an economical development and minimising haulage of imported fill or spoil to and from the development site. Bulk haulage should always be considered an adverse effect on adjacent development, and infrastructure. ***Minimal Road Haulage***

D6.05 SPECIAL TREATMENT OF PARTICULAR AREAS

1. Areas abutting the 100 year ARI flood levels shall be site regraded to a minimum level of 0.5 metres above the 100 year ARI flood levels. In doing so, the Designer shall ensure that other areas are then not affected by flooding. The site shall be identified on the Drawings with appropriate notation of site specific requirements. ***Flooding***

2. In the event that an area is known to be affected by or inundated by local stormwater flows, the Designer shall investigate the existing conditions as they relate to the proposed development and advise the Developer in the preliminary design report on all data obtained in the investigation and recommend appropriate contour adjustments. The report should normally be accompanied by sketch plans to clarify recommendations. ***Inundation Areas***

3. Site constraints either natural or otherwise may be required to be identified as a burden on developed property. It is recommended that the Designer take this into account when preparing the design. The property may ultimately be affected by a "restriction as to user", which may be controlled by a legal 88B Instrument placed on title to the land and/or by a Section 149 message advising prospective purchasers of any restrictions affecting the land.

Restrictions on Land Use

4. The finished surface of filled areas shall be designed to levels allowing an adequate cover depth over the pipeline (if piped) and permitting surface stormwater flow to be guided to inlet pits if depressions are retained in the finished surface contouring.

Piped Gullies or Depressions

5. The location of such features shall be clearly defined on the site regrading plans and defined by distance to corner boundaries, monuments, etc for purposes of relocation at the geotechnical testing stage for work as executed Drawings. A geotechnical report specifying the site specific preparation and compaction requirements will be required to be incorporated with the site regrading plan. A description of the minimum acceptable quality of the fill shall also be specified on the plans, supported by geotechnical recommendations. All documentation necessary from various authorities to support the filling of dams and watercourses shall be supplied with the Drawings.

Dams and Water Courses

6. The finished level of any building area shall be designed to ensure a desirable surface grading of 1.5% (1% minimum) oriented in the direction of the drainage system designed to cater for its catchment.

Flat Ground

7. Building areas containing natural ground slopes of an excessively steep nature, ie greater than 15%, shall be brought to the attention of a Geotechnical Engineer for investigation of compatibility with dwelling types proposed. Specific requirements shall be noted on the Drawings. Residential lots greater than 20% shall have levelled building platforms constructed with geotechnical landslide risk management assessment and demonstrate vehicle access can comply with Councils Vehicular access Design Specification D13. Residential lots should not have grades greater than 25% and will only be accepted on an individual basis with justification.

Steep Slopes

8. In known salt affected areas, or areas found to be salt affected by the geotechnical investigations, the Designer shall evaluate the existing conditions as they relate to the proposed development. The Designer shall also take advice from the relevant land and water resource authority and advise the Developer, in the preliminary design report, of areas requiring action to prevent salinity development. Appropriate regrading strategies aimed at lowering the groundwater table should also be included in the preliminary design report together with primary measures to prevent extension of salinity problems.

Salinity Prevention

D6.06 GENERAL STANDARD OF LOT PREPARATION

1. Special requirements will apply where necessary but generally lots are to be cleared of low scrub, fallen timber, debris, stumps, large rocks and any trees which in the opinion of Council are approaching the end of their functional life or are dangerous or will be hazardous to normal use of the development. Prior consultation with Council's Management Officer is necessary. Such requirements shall be shown on the Drawings.

Clearing

2. All timber and other materials cleared from lots shall be removed from the site unless otherwise required. All roots, loose timber, etc which may contribute to drain blockage shall be removed. Such requirements shall be shown on the Drawings.

Disposal

3. In areas to be filled over butts of trees, allowance is to be made for clearing of all trees and replanting with a minimum of six (6) advanced suitable species to each lot; planting to be clear of probable future building location, and not to be commenced until filling has been completed and graded, with provision for watering and maintenance for duration of the contract. These specific requirements shall be shown on the Drawings.

Overfilling Area of Trees

4. Selected trees shall be preserved by approved means to prevent destruction

Preservation

normally caused by placement of conventional filling or other action within the tree drip zone. The Tree Management Officer shall be consulted for advice and all specific requirements noted on the Drawings. ***of Trees***

D6.07 STANDARD OF FILL FOR LOTS

1. The following notations are to be incorporated in the Drawings. "Filling is to be of sound clean material, reasonable standard and free from large rock, stumps, organic matter and other debris." "Placing of filling on the prepared areas shall not commence until the authority to do so has been obtained from the Council". ***Drawing Notations***

2. All work shall be in accordance with AS 3798. Fill is to be placed in layers not exceeding 150mm compacted thickness. All fill is to be compacted to 95% standard maximum dry density. Maximum particle size shall be 2/3 of the layer thickness. ***Fill Quality***

3. Fill comprising natural sands or industrial wastes or by-products may only be used after the material type and location for its use is approved by Council and will be subject to specific requirements determined by prevailing conditions. ***Restricted Fill***

4. It is essential that prior advice be given of intended use of such materials. It should be noted that failure to obtain Council's approval may lead to an order for removal of any material considered by Council or other relevant authorities as unsuitable or in any way unfit for filling. ***Prior Approval***

5. All areas where filling has been placed are to be dressed with clean arable topsoil, fertilised and sown with suitable grasses. This work shall be carried out in accordance with the Construction Specification for LANDSCAPING – Version 1. ***Top Dressing***

D6.08 TEMPORARY DIVERSION DRAINS

1. Where temporary drains are required to divert surface flows away from the site regrading area, the location and silt/erosion control treatment shall be clearly identified on the Drawings. The scale of such works shall reflect the volume of water to be diverted. ***Silt/Erosion Control***

The objective will be to ensure minimal soil disturbances and material loss off the site.

Control measures will include, but not be limited to:

(a) Provision of trench stops every 30m along a trench, with provision for overtopping to be directed to the kerb.

(b) Placement of "blue metal" bags along kerb and gutter at maximum 30m spacings.

(c) Placement of "blue metal" bags around downstream drainage pits.

The requirements identified in the Design Specification for EROSION CONTROL AND STORMWATER MANAGEMENT – D7 should be addressed for any additional requirements.

D6.09 CONCURRENCE WITH THE ENVIRONMENTAL PROTECTION AUTHORITY (EPA)

1. The Designer is recommended to refer to relevant guidelines issued by EPA with regard to any items requiring specific consideration when preparing a site regrading plan. Such plans may need to incorporate sediment/siltation/erosion/salinity control devices with specific reference to the stage at which these are to be provided. The responsibility shall rest with the Designer/ Developer to make enquiries with EPA and subsequently obtain Council approval to proposed measures. ***Specific Considerations***

D6.10 WORK AS EXECUTED DRAWINGS

1. The Designer shall annotate on the site regrading plan, the site specific detail to be shown on the Work-as-Executed Drawings. Such detail shall include geotechnical report certifying the works to be suitable for the intended purpose and any other certifications, testing and survey data, as required in this Specification.

Site Specific Details

D6.11 CARTAGE OF SOIL

1. The Designer shall refer to Council for acceptable haul roads with applicable load limits. This detail shall be required to be shown on the site regrading plan. The payment of a Bond may be required by the Developer/Contractor where Council has some concern about the ability of a haul road to sustain the loads without undue damage or maintenance requirements.

Possible Bond Requirement

2. Unless specific application is made to Council and approval obtained, the plans will be annotated as follows:

Topsoil

> "All topsoil shall be retained on the development site and utilised effectively to encourage appropriate revegetation."

D6.12 EFFECT ON ADJOINING PROPERTIES

1. Where it is proposed to divert or direct piped stormwater into adjoining properties, drainage easement rights are to be created over the adjoining lots in accordance with the Specification for STORMWATER DRAINAGE DESIGN – D5.

Stormwater Easement

2. A written agreement shall also be sought to carry out construction work on adjoining properties and all such agreements are to be submitted to Council. It should be noted that such works will require development consent and where such consent was not obtained as part of the subdivision approval a further development application for the works will be required to be submitted and approved prior to the commencement of these external works.

Construction Agreement

D6.13 BATTER SLOPES

1. The steepest slope that is permitted on any residential allotment is 1(v):4(h). Any batters that exceed this slope are to be suitably retained with retaining walls (refer to D6.14 RETAINING WALLS).

Requirement for retaining walls

2. The top of batters are to be provided with a level hinge that extends 500mm into the lower allotment as shown in Figure D6.1

Treatment of batter slopes

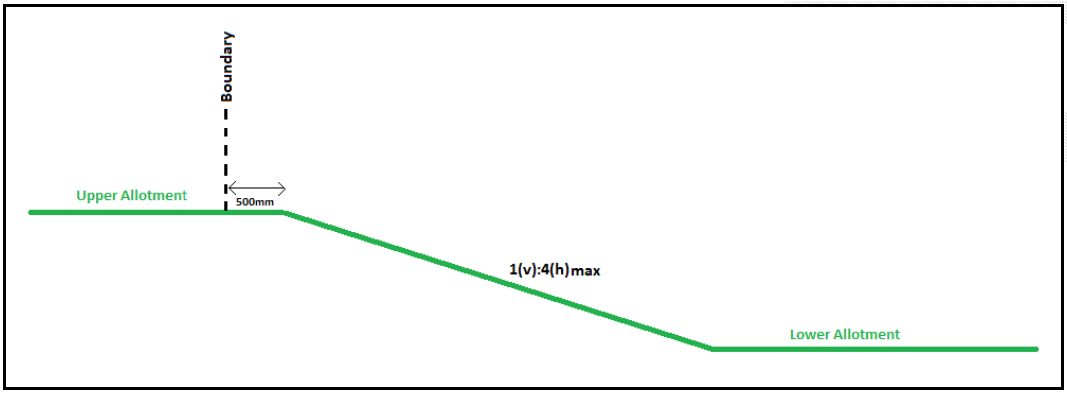

Figure D6.1 – Batter Hinge

3. All batter slopes are to be dressed with clean arable topsoil, fertilised and sown with suitable grasses. This work shall be carried out in accordance with Council's Construction Specification for LANDSCAPING.

D6.14 RETAINING WALLS

1. Where batter slopes on residential allotments exceed 1(v):4(h) they are to be suitably retained with retaining walls. ***Requirement for retaining wall Certification***

2. Retaining walls that exceed 1m in height are to be certified by a structural engineer.

3. Individual retaining walls are to be a maximum of 1.5m in height. ***Height of Retaining Walls***

4. Retaining walls are to be located such that the rear face of the wall and any associated footings are located wholly within the boundary of the lower allotment and on the lower side of the 500mm hinge as shown in Figure D6.2. ***Siting of Retaining walls***

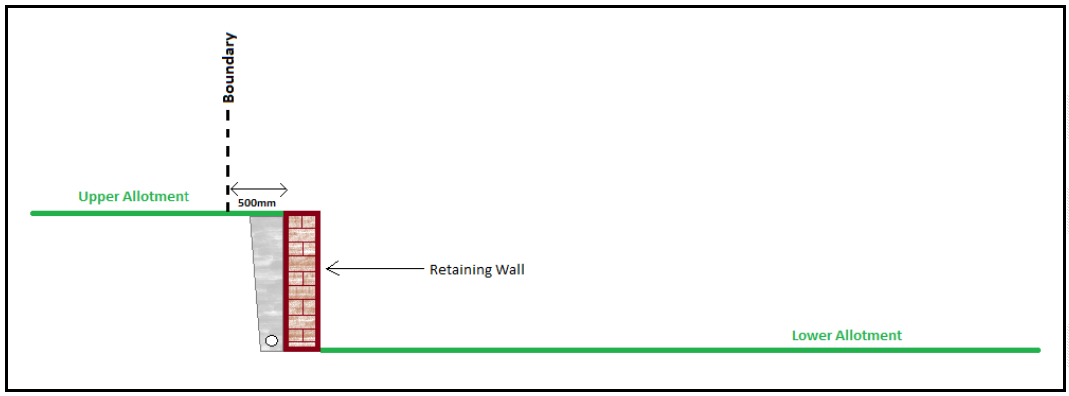

Figure D6.2 – Siting of Retaining Walls

SPECIAL REQUIREMENTS

D6.15 RESERVED

D6.16 RESERVED

D6.17 RESERVED

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